

Line Maintenance Operational Dashboard

Live app: <https://jfk-ops-live-dashboard-vercel-mirro.vercel.app/>

Overview

The Line Maintenance Operational Dashboard is a low-code, highly tailorable web app designed to support daily line maintenance execution, resource allocation, and shift-level decision making. It gives engineers and managers one live operational view that combines daily roster context, live flight activity, weather visibility, and assignment control.

What it does

- **Daily Roster CSV Upload:** imports station activity and creates a practical operational board for the day.
- **Interpretative ingestion logic:** intelligently parses roster content and proposes assignment mappings, improving compatibility across different roster formats.
- **Real-time dashboard access:** engineers can view the operational picture on the go (flight rows, status, gate, reg/type where available).
- **Live weather panel:** station weather now/AM/PM/EVE with precipitation to support tactical planning.
- **Resource assignment layer:** certifier/mechanic assignment controls with quick edit and reset behaviors for fast replanning.
- **Gantt operations view:** managers can see the day timeline, overlapping workload, and peak periods at a glance.
- **Heat-map style demand insight:** highlights busy intervals to support staffing and shift pacing decisions.

Enrichment & resilience layers

- Flight enrichment stack designed with fallback behavior (provider + cache + manual overrides).
- A/C registration and type population can be sourced from live/enrichment layers where available.
- DOC8643 type normalization support helps standardize aircraft type values into cleaner outputs.
- Manual column edits remain available for operational control when feeds are incomplete.

Why this is valuable

- **Resource management:** aligns people, flights, and constraints in one execution surface.
- **Faster decisions:** combines planning data and live cues into a single operational picture.
- **Station versatility:** adaptable by station and process maturity without expensive enterprise customization.
- **Low-cost deployment:** built as a practical low-code web application with minimal overhead.

Screenshots

Live tracker temporarily unavailable; using fallback snapshot + schedule.

Airline	Flight	A/C Reg	A/C Type	Terminal	Gate	ETA	STD	Status	Certifier	Mechanic	Notes
JL	JL006/005	-	A350	8	-	1100	1350	Scheduled	Andrew Ibsen	Assign...	-
NH	NH110/109	-	B777	7	-	1130	1405	Scheduled	Andrew Ibsen	Assign...	-
BA	BA117/176	-	B787	8	18C	1217	2110	Scheduled	Andrew Ibsen	Nikolas Dundon	-
BA	BA175/172	-	B787	8	18D	1334	2140	Scheduled	Andrew Ibsen	Nikolas Dundon	-
EI	EI105/104	-	A330	7	-	1455	1800	Scheduled	Rahman Arian	Fahim Abrar	-
BA	BA173/112	-	B787	8	20	1521	1915	Scheduled	Devran Turegun	Nikolas Dundon	-
QF	QF3/4	-	B787	8	-	1525	1815	Scheduled	Ray Abes	Naresh Dindiall	-
EI	EI111/110	-	A321	7	-	1625	1830	Scheduled	Mark Ferrel	Fahim Abrar	-
IB	IB211/212	EC-OIL	A32X	8	-	1625	1755	Scheduled	Saleh Al Assaf	Assign...	-
ZO	Z0701/702	G-CKWP	B787	7	-	1655	1905	Scheduled	Assign...	Assign...	-
BA	BA177/174	-	B787	8	16	1658	1950	Scheduled	Rahman Arian	Frank Richmond	-
NZ	NZ2/1	-	B787	1	-	1700	1920	Scheduled	Assign...	Assign...	-

Figure 1: Daily roster board with assignment columns, live statuses, and editable operational controls.

Documentation & Examples

Everything you need to integrate with ADS-B Exchange

Quick Start Examples

JavaScript

Python

cURL

ADS-B Exchange Javascript Example

JAVASCRIPT

```
// ADS-B Exchange API Example
const response = await fetch('https://www.adsbexchange.com/data/', {
  method: 'GET',
  headers: {
    'Content-Type': 'application/json'
  }
});

const data = await response.json();
console.log(data);
```

Copy

Unlock the power of the ADS-B Exchange API, which provides seamless access to comprehensive real-time and historical data of airborne aircraft. With a robust dataset that covers a wide array of vital flight information, this API is designed for developers, aviation enthusiasts, and organizations requiring detailed aerial insights. Whether you are building flight tracking applications, conducting aviation research, or simply eager to stay informed about ongoing air traffic, the ADS-B Exchange API delivers a reliable solution to tap into an extensive pool of aircraft data, ensuring you stay updated with the latest airborne activity at your fingertips.

Utilizing the ADS-B Exchange API comes with numerous advantages. The key benefits include:

Real-time tracking of active flights, enhancing situational awareness.

Access to historical flight data for comprehensive analysis.

Coverage of a vast number of aircraft worldwide, ensuring extensive data availability.

User-friendly documentation and support, allowing for quick integration.

Free access to open-source data, promoting transparency and collaboration in aviation information sharing.

Figure 2: Live ops context showing weather module, feed health, and roster execution status.